

Active Leash for Intelligent Vigilance & Ease A.L.I.V.E.

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Abstract—Running with a dog is meant to be a shared and enjoyable experience. However, sudden changes in the dog’s speed can generate abrupt leash tensions, reducing comfort and safety for the runner. This project presents the design, modeling, and experimental evaluation of an active dog leash capable of assisting the runner by regulating leash tension and mitigating sudden accelerations. Experimental tests demonstrate the ability of the prototype to reduce peak forces and smooth leash dynamics, improving stability during running, within its constraints and range of application. The results highlight the potential of active assistance devices to enhance safety and comfort for both dog and owner.

Index Terms—Active leash, wearable robotics, human–animal interaction, motion control, embedded systems, assistive device, safety in locomotion.

I. INTRODUCTION

JOGGING with a dog is commonly associated with physical activity, well-being, and strengthened human–animal bonding. However, this activity can also present significant safety risks when the dog suddenly pulls or tightens the leash, potentially causing the owner to lose balance. Such incidents are not uncommon and may lead to serious injuries. Between 2001 and 2018, more than 350,000 dog leash–related injuries were treated in United States emergency departments, corresponding to approximately 20,000 cases per year [1]. According to previous studies [2]–[4], approximately 54% of these injuries result from sudden pulling by the dog, while 38% occur when the leash becomes tangled, causing the owner to trip and fall. These numbers represent ER-reported cases only, and likely underestimate the total burden because many people are treated at home.



Fig. 1. Examples of conventional dog leashes: a rope leash (a), a retractable leash (b), and an elastic leash (c).

Having identified these two primary injury mechanisms, it is relevant to assess whether existing leash designs adequately address these risks. The most basic solution available on the

market is the standard fixed-length leash (a) [5]. This type of leash is essentially a rope that provides no distance regulation, shock absorption, or retraction mechanism. As a result, it neither prevents sudden pulling forces nor reduces the risk of the leash dragging on the ground.

A second widely used solution is the passive retractable leash (b) [6], which employs a rotary spring mechanism to automatically retract the leash. This design reduces the likelihood of leash dragging and therefore lowers the risk of tripping. Additionally, such leashes typically offer two operating modes: a locked mode, where the leash behaves like a conventional fixed leash, and a free mode, where the dog can extend the leash smoothly while moving away from the owner. Although this configuration reduces the perception of sudden pulling during extension, a significant force is still transmitted to the owner once the leash reaches its maximum length, potentially resulting in destabilizing shocks.

A third existing solution is the elastic leash (c) [7], which introduces a degree of compliance to absorb part of the pulling force. While this approach improves comfort compared to rigid designs, it remains passive and does not actively regulate the distance or force applied between the dog and the owner.



Fig. 2. ALIVE's prototype worn by a user to illustrate size and ergonomics.

To address the limitations of current solutions, we propose ALIVE, an active leash specifically designed for dynamic activities such as jogging and walking. ALIVE incorporates an adaptive tension control system that continuously regulates the distance between the dog and the owner while actively absorbing variations in speed and force. By minimizing sudden jerks and preventing leash slack, the system enhances stability and safety for the user while preserving freedom of movement

for the dog. This design aims to reduce injury risk and improve the overall experience of shared physical activity between humans and their canine companions.

II. ENGINEERING SPECIFICATIONS

A. Principal Function

ALIVE's principal function is to reduce sudden leash tension peaks transmitted to the user (jerk/shock when the dog accelerates) while also maintaining a safe, comfortable reference distance between dog and user. In practice, this means (i) impulsive pulling forces need to be damped, (ii) leash slack that could cause tripping should be prevented, and (iii) the dog should remain within a user-defined distance during motion.

B. System Definition

The system consists of:

- 1) A winding mechanism to wind and unwind the rope, managing leash length.
- 2) A braking mechanism to resist excessive extension by preventing further unwinding of the leash and producing a damping effect.

C. Step-by-Step Working Principle

- 1) The user sets a desired leash length L_{ref} (target distance, e.g., 1–2 m).
- 2) The system continuously estimates the current leash length L .
- 3) If the dog stays within the target range ($L \leq L_{\text{ref}}$), the system keeps the leash taut not dragging on the ground
- 4) If the dog exceeds the target length ($L > L_{\text{ref}}$), the system activates controlled braking to limit further extension and reduce jerk.
- 5) When the dog is stopped, braking is released and then the dog is pulled back by the leash within the desired distance.

This creates a leash behavior that is stable and controlled compared with a regular leash.

D. Motivation

The motivation is user safety and comfort. With ALIVE, the user is able to set a desired distance between them and the dog while also reducing sudden force peaks if the dog exceeds that distance.

E. Engineering Requirements

As shown in Table I, the design targets are quantified as:

- **Damping range: 50%.** The system should reduce the peak pulling force felt by the user by at least 50% compared to a regular leash during a sudden pull event.
- **Dog specifications: all types of dogs.** The design should be scalable across a wide range of dog sizes and strengths.
- **Relative speed: 5 m/s.** The system should remain functional under sudden and fast pulls.

- **Braking force: 400 N.** To choose a reasonable peak braking force, we consider the following calculations. If a dog of mass m reaches a relative speed v and the leash/brake stops this motion over an extra extension distance ΔL , then the work done by the brake roughly matches the kinetic energy:

- Kinetic energy: $E = \frac{1}{2}mv^2$
- Work by braking force: $W \approx F \Delta L$

Therefore:

$$F_{\text{avg}} \approx \frac{\frac{1}{2}mv^2}{\Delta L} \quad (1)$$

We pick conservative but realistic values:

- $m = 25$ kg (medium/large dog)
- $v = 5$ m/s (chosen relative speed)
- $\Delta L = 1$ m (allowed extra extension while braking)
- safety factor $SF = 1.3$ as an engineering margin to account for uncertainties

Calculation:

- $E = 0.5 \times 25 \times 5^2 = 0.5 \times 25 \times 25 = 312.5$ J
- $F_{\text{avg}} = 312.5/1 = 312.5$ N
- With safety factor: $F_{\text{max}} \approx 1.3 \times 312.5 \approx 406$ N

So we round to a clean design target: ~ 400 N peak braking capacity at the leash.

- **Winding force: up to 150 N peak**

To select a reasonable peak winding force, we relate it to the dog's ability to resist a horizontal pull through ground traction. On flat ground, the maximum friction (traction) force is approximately

$$F_{\text{max}} \approx \mu N \approx \mu mg,$$

where μ is the ground friction coefficient and $N \approx mg$ is the normal force. Using the same design dog mass as in the braking estimate ($m = 25$ kg) and a typical value $\mu \approx 0.6$, we get

$$F_{\text{wind,peak}} \approx 0.6 \times 25 \times 9.81 \approx 147 \text{ N} \approx 150 \text{ N}.$$

Therefore, a peak winding force of about 150 N is chosen as a reasonable pull-back capability while remaining below the peak braking target (400 N). This is chosen especially for both the user and dog's safety.

- **Reference distance: 1–2 m.** The user must be able to set the desired leash length within 1 to 2 m, and the system should regulate around that value.

TABLE I
ENGINEERING SPECIFICATIONS

Specification	Target
Damping range	50%
Dog specifications	All types of dogs
Winding force	150 N
Braking force	400 N
Relative speed	5 m/s
Distance of reference	1–2 m

F. Verification Questions

To check whether a design meets these specifications, we will evaluate the following engineering questions:

- 1) **Damping requirement:** Does the system reduce peak pulling force by $\geq 50\%$?

$$\frac{F_{\text{peak,system}}}{F_{\text{peak,baseline}}} \leq 0.5 \quad (2)$$

- 2) **Distance regulation:** When the user sets L_{ref} (1–2 m), does the leash length remain near this range during motion, without excessive slack or uncontrolled extension?
- 3) **Braking capability:** Can the brake provide the required resisting effect when $L > L_{\text{ref}}$?
- 4) **Dynamic condition:** Under fast pulling (relative speed up to 5 m/s), does the system remain safe and controlled (no excessive jerk)?
- 5) **Robustness:** Does the system concept remain valid for different dog sizes/strengths?

III. METHODS

A. Measurement of Proposed Design's Solution

In order to verify that the prototype satisfies the objectives defined in the engineering specification table, measurement methods were selected to directly target the critical functional requirements of the system. The primary objective of the prototype is to damp sudden pulls and reduce the peak force transmitted to the user. Consequently, force amplitude was selected as the primary performance metric, as it provides a direct and quantitative indicator of the system's effectiveness in terms of user comfort and safety.

Since sudden pulls generate short-duration force spikes, the measurement strategy focuses on capturing transient peak forces rather than static or average forces. These peaks represent the most critical operating condition for the user and therefore constitute the most relevant quantity for evaluating the damping performance of the system. The experimental setup described in section III-C was specifically designed to enable the measurement of these transient events.

Force measurements were performed using a dynamometer combined with a camera-based tracking method. A digital dynamometer alone was insufficient to reliably capture rapid force variations. Video recording allowed force data to be extracted using a tracking algorithm, ensuring that short-duration force peaks were accurately quantified. This approach enables the measurement of the actual force experienced by a user when the dog, modeled as a moving weight, pulls away abruptly under different scenarios.

In addition to force, the leash length over time was measured as a secondary performance metric. This measurement was selected to evaluate the dynamic behavior of the system and its ability to regulate the dog's motion. Monitoring leash length allows verification that the leash extends when the dog moves away, stabilizes when the system applies braking or control, and retracts when the dog is slowed down and brought back. Together, force and leash length measurements provide complementary information on both the damping performance and the functional behavior of the prototype.

B. Validation of Design

The validation of the proposed design is based on comparing measured quantities across different configurations to assess whether the system meets the performance objectives defined in the engineering specifications. According to these specifications, the design is considered valid if the peak force transmitted to the user during a sudden pull is reduced by at least 50%.

To isolate the contribution of each design element, several configurations were tested. A single rope without any control was used as the reference case, representing the maximum force transmitted to the user. This configuration was compared to a rope equipped with a passive spring to evaluate whether passive damping alone is sufficient to significantly reduce force peaks. Finally, the uncontrolled rope was compared to the same rope equipped with the active control provided by the prototype.

The peak forces measured in each configuration were extracted and compared. The ratio between the peak force in the controlled configuration and the peak force in the reference configuration directly quantifies the damping efficiency of the system. Achieving a reduction of at least 50% in peak force demonstrates compliance with the primary design objective and validates the effectiveness of the proposed solution.

Leash length measurements further support the validation of the design by demonstrating correct system behavior over time. An increase in leash length confirms that the system allows the dog to move away, while a stabilization of the leash length indicates effective braking and force damping. The subsequent decrease in leash length when the dog is slowed down and brought back confirms that the system maintains control without abrupt force transmission. This behavior confirms that the prototype not only reduces peak forces but also ensures smooth and controlled interaction, thereby validating the overall design.

C. Experimental Setup

To validate the performance of the proposed system, several experiments were conducted:

- 1) evaluation of an additional spring at the end of the leash providing increased damping in it;
- 2) comparison between controlled, uncontrolled, and conventional leashes;
- 3) comparison between the real-time feedback position and the theoretical position.

Fig. 3 and 4 show the experimental set up to conduct our experiments. Experiment #1 was designed to assess the effect of an additional spring under a sudden pull. A 1 kg mass was released from a fixed height, while the dynamometer scale was recorded to measure the resulting force. The rope length was kept constant, and the initial elongation of the added spring at the tip was taken into account, as shown in Fig. 3.

Experiment #2 was designed to evaluate inertia reduction using our prototype under sudden pulls. A rolling kart ($m_{\text{kart}} = 1.243 \text{ kg}$) was released from a fixed height, while the dynamometer scale was recorded to measure the resulting force. Additionally, the position was recorded with a ruler to

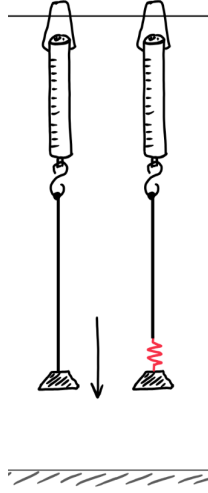


Fig. 3. Set up used for Experiment #1. A mass was attached to a rope, with an optional spring connected in series.

provide reference data for Experiment #3 (control condition). The spring was mounted in series at the end of the rope, and their initial total length was kept constant for each scenario.

Since the maximal acceleration supported by the main motor is $a_{max} \approx 3.0 \text{ m/s}^2$, we created a slope to simulate this acceleration using gravitational forces. The angle needed was found by following calculations:

$$a_{max} = g \sin(\theta) \Rightarrow \theta = 20^\circ$$

Experiment #3 was conducted to demonstrate that the system operates as intended, allowing the dog to move freely within the free range and, once this range is exceeded, slowing the dog down and guiding it back inside the free range.

Experiment #4 was conducted to assess the accuracy of the controller by comparing the actual leash position with the motor position.

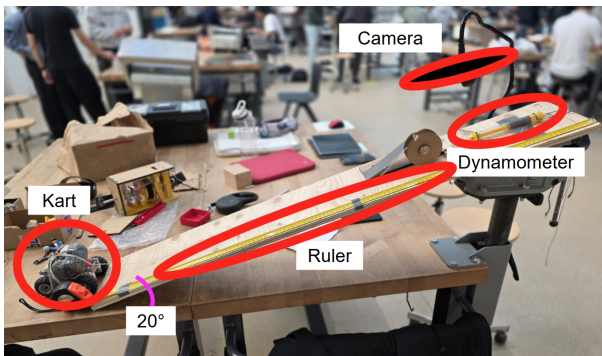


Fig. 4. Set up used for Experiment #2, #3 and #4, including the kart on an inclined plane, the ruler for displacement measurement, the dynamometer for force measurement, and the camera for motion tracking.

IV. SOLUTION OPTIONS

A. Initial solution: Railing system

An initial railing system was developed using two actuators: one controlling the spool winding and the other driving a pulley cart that guides the leash toward the dog (Fig. 5). The

pulley cart moves laterally along a belt (Fig. 6) through a toothed belt transmission connected to the actuator. The belt is composed of articulated straight segments incorporating a rigid rail that constrains and guides the motion of the pulley cart. This modular design was intended to allow the rail to be adjusted to different user waist geometries.

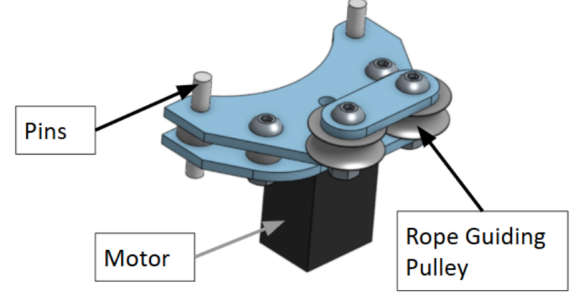


Fig. 5. Pulley cart guiding the rope along the belt. This component is actuated by the secondary motor.

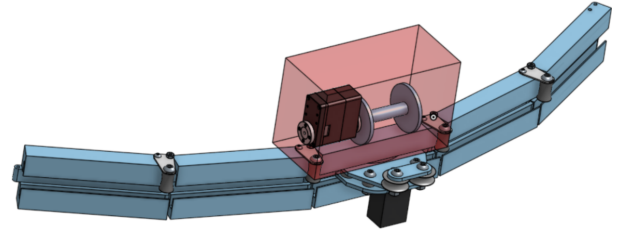


Fig. 6. Assembled belt with winding box (red) and pulley cart

This idea required a motor capable of winding and unwinding the leash according to the dog's movements. This actuator needed to be sufficiently fast to avoid constraining the dog when it runs away, allowing free movement, while also being strong enough to bring the dog back within a specified range when this limit is exceeded. We therefore searched for a motor providing both high torque and high speed within our budget. However, this requirement proved to be unrealistic, as a trade-off between speed and torque is generally unavoidable. Fig. 10 shows that this trade-off is not feasible for this system.

Different options for the selection of sensors and actuators were investigated to determine the most suitable configuration for the initial prototype. These options are summarized and compared in Table II. Solution "Embedded Sensor" was chosen initially due to its compactness, better motor performance, and wiring simplicity.

At this stage, we reached a turning point and had to rethink the mechanism and introduced a braking system, changing the original DC motor to another high torque servo, ST3020.

B. Working Principle of Proposed Design

ALIVE integrates two main functional subsystems within a single casing: a winding system responsible for maintaining leash tension and retracting the leash to a predefined length, and a braking system that limits the dog's speed when operating beyond the target distance.

TABLE II
COMPARISON OF SENSORS AND ACTUATORS SELECTION FOR THE INITIAL RAILING SYSTEM.

Engineering specifications	Solution "Stretch sensor"	Solution "IMU"	Solution "Embedded sensor"
Composition	Servo ST3020 + DC motor Stretch sensor on leash Potentiometer & Switch	Servo ST3020 + DC motor Load cell GPS + IMU (Human speed)	Servo ST3025 + DC motor Load Feedback given by servo Potentiometer & Switch
Range of motion	Servo: 360° (full rotation) Stretch sensor on the leash	Servo: 360° (full rotation) GPS: unlimited IMU: +/- 2g to +/- 16g (sensitivity)	Servo: 360° (full rotation) In-built load feedbacks
Torque	Servo: 25 kg · cm	Servo: 25 kg · cm	Servo: 40 kg · cm
Bandwidth rpm	60 rpm (12V)	60 rpm (12V)	85 rpm (12V)
Price	Servo: 26 CHF Sensors: < 35 CHF	Servo: 26 CHF Sensors: < 60 CHF	Servo: 90 CHF Sensors: < 10 CHF
Drawbacks	- Wiring from user to the dog.	- Implementation of the GPS/IMU & accuracy.	- Load feedback needs filtering.
Advantages	+ Knowledge of tension in the leash.	+ Speed difference between the user and the dog	+ Compact design

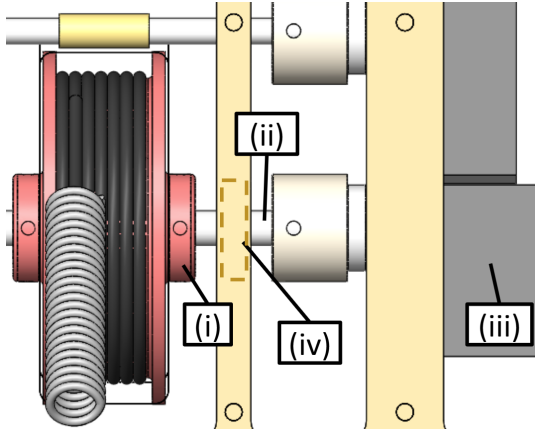


Fig. 7. The winding system, composed of the main spool (i), a steel axis (ii), the primary servomotor (iii) and bearings (iv).

1) *Winding system*: The winding system showed in Fig. 7 consists of a main spool mounted on a steel shaft, driven by the primary servomotor, and supported by bearings to ensure smooth rotation. It is responsible for controlling the dog's distance by winding and unwinding the leash as required. The system maintains continuous leash tension to prevent ground contact, thereby reducing the risk of injury. In addition, it is designed to retract the dog when the predefined leash length is exceeded.

2) *Braking system*: The braking system shown in Fig. 8 consists of a brake disc mounted on the same steel shaft as the spool, brake pads attached to actuation levers, and a left- and right-hand threaded shaft driven by a secondary servomotor. The threaded shaft converts rotational motion into linear motion, actuating the levers and causing the brake pads to clamp the disc and generate braking torque. The shaft is supported by bearings to ensure smooth rotation. The design is inspired by bicycle braking systems, which are known for their high braking efficiency.

3) *Basic Calculation for Solution Validity*: The system, shown in Fig. 9, converts input torque M_1 into output braking torque M_2 . Using three system equations, we derived the relationship between input and output torque (3). Calculations

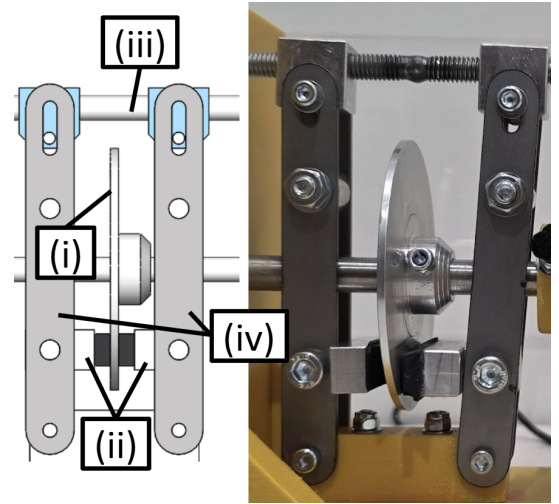


Fig. 8. The braking system, composed of the brake disc (i), brake pads (ii), a threaded shaft (iii), and actuating levers (iv).

with our actual system values show that the output torque is approximately 12 times greater than the input torque (4).

$$\begin{cases} M_2 = 2\mu R F_2 \\ L_1 F_1 = L_2 F_2 \\ M_1 = \frac{p}{\pi} \eta F_1 \end{cases} \Rightarrow M_2 = \frac{2\pi\mu L_1 R \eta}{L_2 p} M_1 \quad (3)$$

$$\Rightarrow \frac{M_2}{M_1} = \frac{2\pi\mu L_1 R \eta}{L_2 p} \approx 12 \quad (4)$$

V. DESIGN ITERATION FROM THE CHOSEN SOLUTION

A. Spool sizing

To be compliant with current leash designs and additional functionalities, ALIVE faces multiple challenges. Both winding speed and torque are required to slow down and pull back the dog at a sufficient rate. However, maximum torque and speed are not proportional, so a trade-off in the mechanical design must be made in order to achieve sufficient

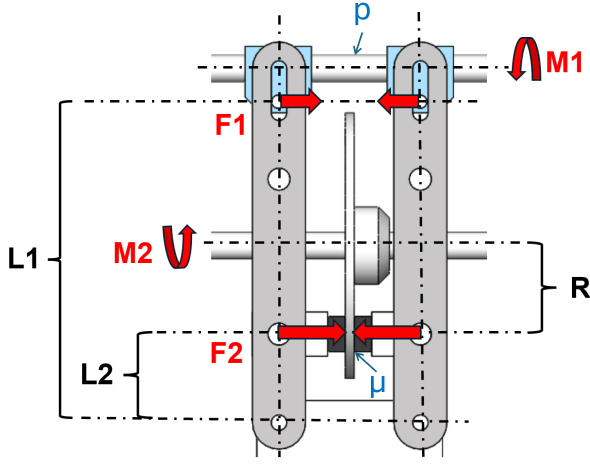


Fig. 9. Characterization of the braking system

TABLE III
SYMBOLS DEFINITION OF BRAKING SYSTEM

Symbol	Description	Value
M_1	Input torque	Variable
F_1	Linear force	$F_1 = f(M_1)$
L_1	Distance of F_1 from the pivot point	77 mm
M_2	Output torque	Variable
F_2	Braking force	$F_2 = f(F_1)$
L_2	Distance of F_2 from the pivot point	59 mm
R	Distance of F_2 from the disc axis	19.6 mm
μ	Friction coefficient	0.25
p	Thread size	1 mm
η	Thread friction efficiency	0.25

speed while maintaining adequate torque for our needs.

We chose a high-torque ST3025 servo motor with built-in sensing feedback for the winding mechanism. Since the motor is geared (and therefore not fully back-drivable), it introduces resistance when the dog pulls, especially at the start of motion. We experimentally characterized the motor response at 80% command. As shown in Figure 11, the spool reaches a maximum speed of approximately 0.25 m/s and a maximum acceleration of around 3 m/s². This defines the motor's practical working range: within these limits, the controller can react and regulate the leash smoothly, while outside this range sudden pulls can still create short tension peaks due to the motor's inertia and transmission resistance.

The spool fixed to the shaft provides control over both the output torque and winding speed by selecting an appropriate spool radius, as shown in Figure 10. Taking into account winding speed, torque, free resistance, and available space constraints, a final spool radius of 35 mm was selected. This configuration provides a winding speed of 0.3 m/s and an available pulling force of 112 N.

To ensure a sufficient and consistent slowing force despite the mechanical trade-off, particularly when the dog accelerates

before reaching the target leash length, we implemented a controlled braking system. This system allows the leash mechanism to efficiently dampen motion and safely stop the dog.

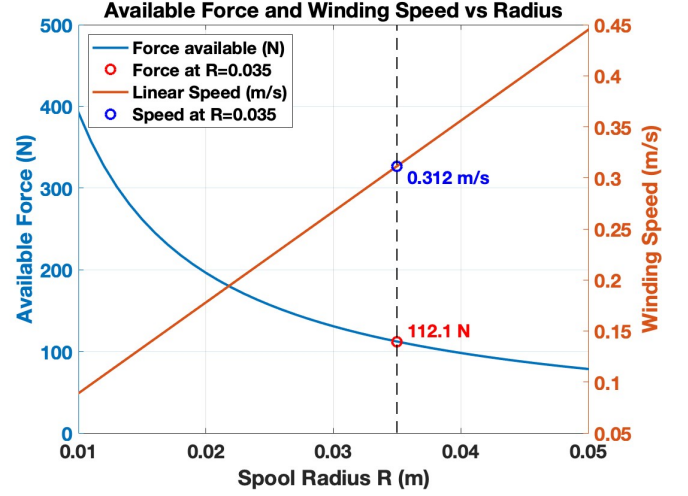
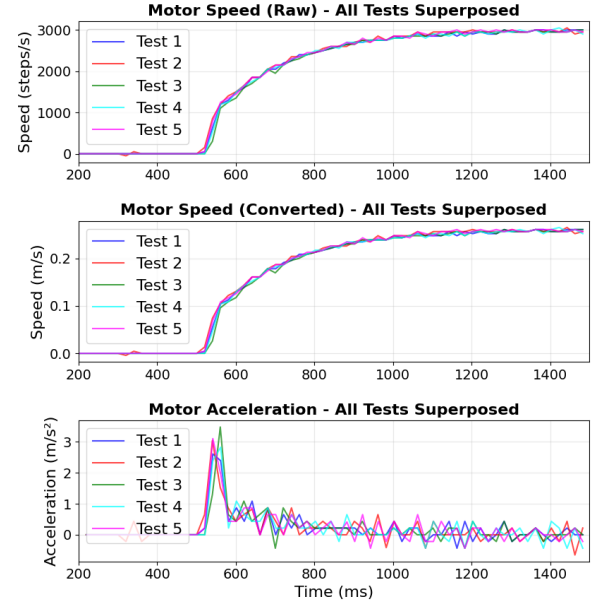


Fig. 10. Winding speed and available force of the primary motor over spool radius

B. Internal friction and residual inertia

Fig. 11. Primary servo motor acceleration and speed limitations at 80% capacity with a spool radius of 3.5 cm. Based on the tests, the maximum speed available is 0.25 m/s and the maximum acceleration is around 3 m/s².

We implemented an inertia controller (see Sections VI-C and VI-D) to further reduce the perceived inertia when the motor operates within its working range (i.e., velocities below 2.5 m/s and accelerations below 3 m/s², see Figure 11).

To counter the residual effects of limited speed response, we added an additional spring at the tip of the leash to provide a damping effect and reduce the sudden pull force felt by both the user and the dog. The spring stiffness introduces an

additional design parameter: reducing the stiffness increases the damping effect during sudden pulls, but as a drawback, it also increases the required elongation (see Figure 12) and add noise to the rope length estimate. To prevent the spring from reaching plastic deformation, we implemented a structural limit that constrains its maximum elongation and avoids plastic deformation or rupture.

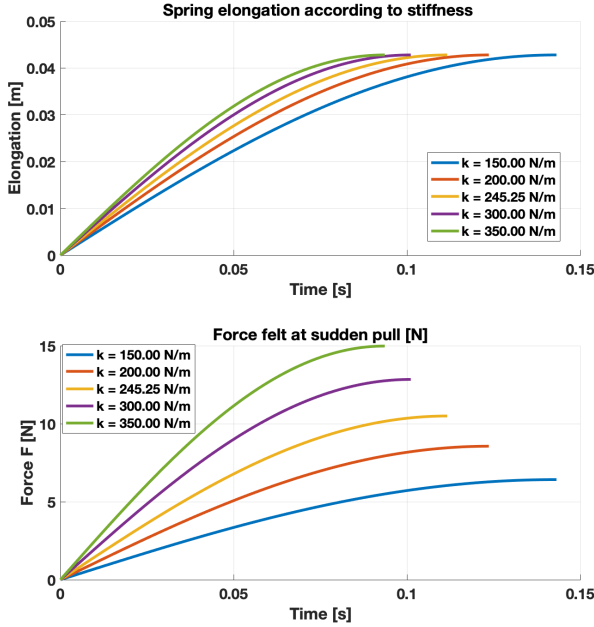


Fig. 12. Spring elongation & Force felt by both user and dog over time at a sudden pull for different spring constants. Analytical solution based on the experimental set up data.

To compute the actual spring constant, a 0.5 kg mass was attached to the spring. An elongation of 2 cm was then measured at the equilibrium:

$$F_{grav} = F_{spring} \quad (5)$$

$$\Leftrightarrow mg = k\Delta x \quad (6)$$

$$\Leftrightarrow k = \frac{mg}{\Delta x} \quad (7)$$

$$= \frac{0.5 \cdot 9.81}{0.02} \quad (8)$$

$$= 245.25 \text{ N/m} \quad (9)$$

Since the spring is contained in a casing, the maximal elongation allowed is 9 cm, leading to $F_{spring,max} = 22.07 \text{ N}$.

VI. FINAL SOLUTION PRESENTATION

A. CAD Model

ALIVE is the product of many iterations of its subsystems. After research and optimization, we came out with the final prototype dimensions.

Fig. 13 shows the final CAD model of the prototype with all validated overall dimensions. Detailed drawings with precise dimensions of the spool and the main components of the

system are provided in the appendix, section X-D, Figs. 28 and 26.

TABLE IV
SUMMARY OF CRITICAL DIMENSIONS

Dimensions	[mm]
Volume	200x90x120
Spool inner radius	35
Braking disk diameter	60
Linear motion lever	86
Braking lever	20
Primary motor torque	40 [kg · cm]
Available force	112 [N]
Winding speed	0.25 [m/s]
Secondary motor torque	30 [kg · cm]
Available braking force	2000 [N]

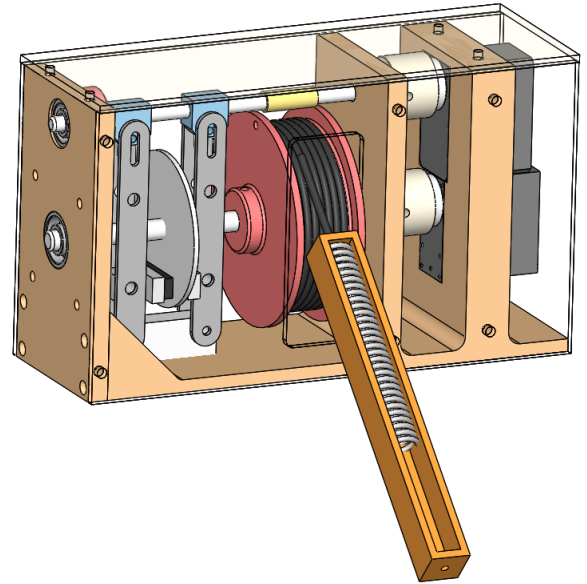


Fig. 13. Overview of the ALIVE system.

B. Electric connectivity

As shown in Fig. 14, the system integrates two actuators, sensors, and control boards to enable active leash functionality. The primary servo (40 kg-cm) drives the winding spool, while the secondary servo (30 kg-cm) actuates the braking mechanism. Feedback is provided by a potentiometer, a push button, and the servos' inbuilt position, speed, and load sensors. Control and communication are handled by an Arduino Mega connected to a Serial Bus Waveshare board.

C. Flowchart of the working principle

The user can calibrate the system using a button to define the zero position of both the rope length and the braking system. The zero position of the braking system is determined automatically by rotating the spool at a constant speed while progressively advancing the braking pads until contact is

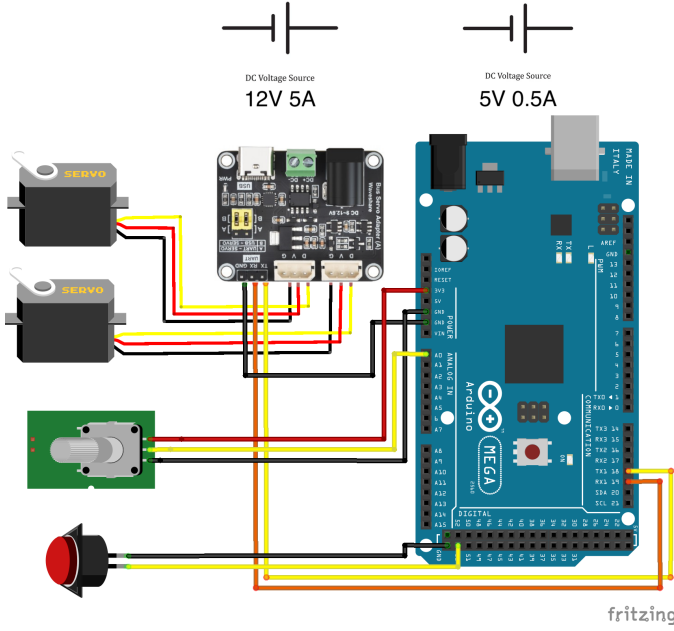


Fig. 14. Overview of the electrical connectivity between actuators, sensors, and controller.

detected, indicated by an increase in the load measured on the spool motor. The braking pads are then retracted by a small, fixed distance, and this position is set as the braking system zero. The zero position of the rope length is set manually by the user. During calibration, the user presses the button to rewind the rope and presses it again once the desired zero position is reached.

A potentiometer is used to define the desired rope length, corresponding to the distance beyond which the system begins to slow down the dog. The user can adjust this distance at any time.

Furthermore, the system uses embedded feedback from the primary (spool) motor to estimate:

- the rope length: the unwrapped step-position feedback of the spool motor, converted to meters using the spool radius;
- the relative speed between the user and the dog: the velocity feedback of the spool motor;
- the load applied by the dog to the user: the load feedback of the spool motor, compensated by subtracting the motor friction model (Fig.15). The initial friction model consisted of a static and a viscous term proportional to the motor speed, with the model coefficients identified experimentally by operating the motor at different speeds and in both rotational directions. However, it was observed that the Stribeck effect was not negligible and that more accurate load values were needed at low speed. Consequently, the friction model was modified to directly map a friction value to each motor speed.

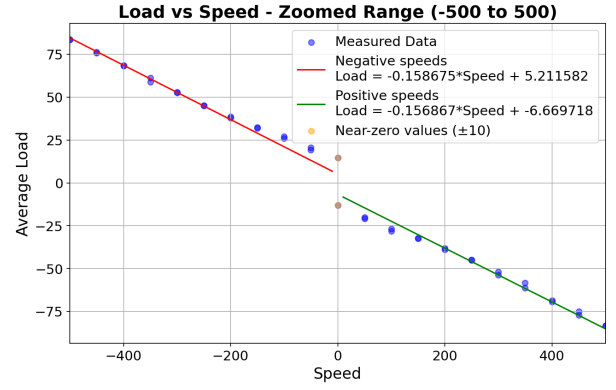


Fig. 15. Friction values of the spool motor, found by making the motor run at fixed speed in both direction with no load applied to it. In motor units.

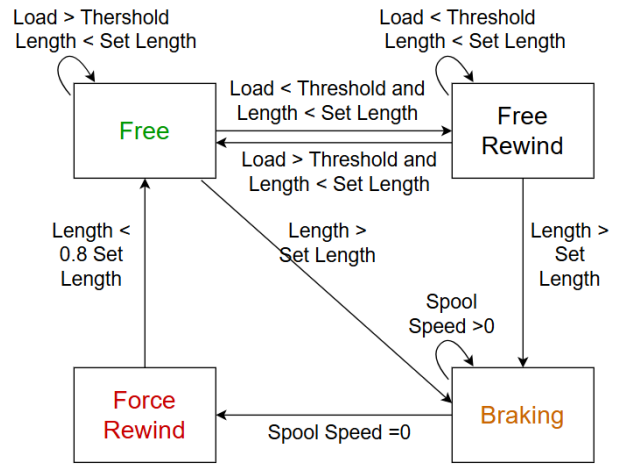


Fig. 16. ALIVE's state machine.

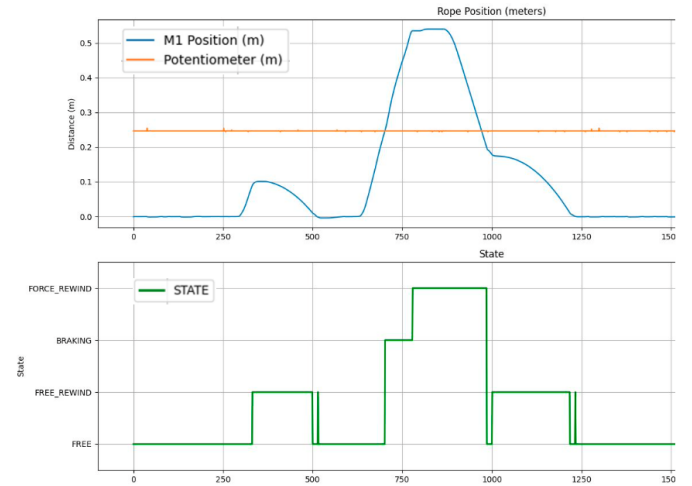


Fig. 17. Leash length over time and its corresponding state. Two pulls were applied before releasing the rope. The first pull ended before the set length (no activation of the braking system) while the second ended after the set length (activation of the braking system).

As illustrated in Figs. 16 and 17, when the rope length remains below the set value, the system switches between the FREE and FREE_REWIND states depending on the tension

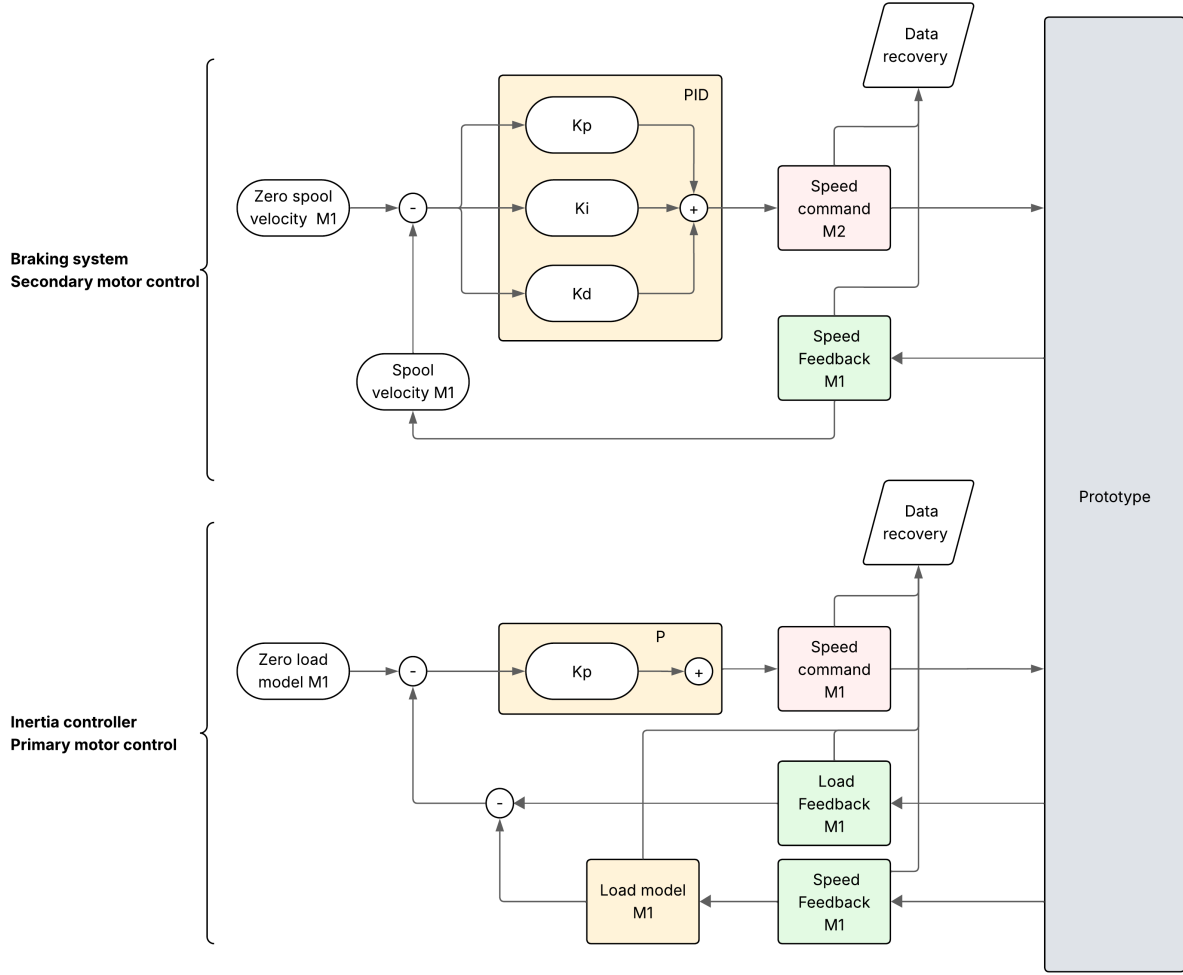


Fig. 18. Slowing system controller and inertia controller

applied to the rope. Once the rope length exceeds the set value, the braking system engages to reduce the relative speed to zero (BRAKING state). The primary motor then maintains the dog's position while the brakes are released, after which the spool motor rewinds the leash until the rope length returns within the prescribed limit (FORCE_REWIND state).

D. Control logic

The closed-loop controllers of the system are shown in Fig. 18.

The first control loop regulates the position of the braking system. It uses the spool motor speed as feedback and compares it to a zero-speed reference. The resulting speed error is processed by a PID controller, whose output is the commanded speed of the brake motor. This controller is activated when in the BRAKING state.

The second control loop is designed to compensate for the spool motor's friction and inertia, making them less perceptible to the user when the rope is pulled (FREE state). This controller uses feedback from the spool motor to estimate motor friction via the friction model. The load applied by the dog to the user is then obtained as the difference between the

measured motor load and the estimated friction. This estimated load is compared to a zero-load reference, and the resulting load error is processed by a proportional controller, whose output defines the speed command for the spool motor. Fig. 19a in the results shows that this controller was effective in reducing the motor's inertia, although it was not yet perfect.

The rope rewind (in both FREE_REWIND and FORCE_REWIND states) is not controlled by a closed-loop system, but is instead time-based: the longer the system remains in this state, the faster the rope rewinds, up to a predefined maximum speed.

VII. RESULTS

This section presents the experimental results obtained with the proposed prototype. The system performance is evaluated through a series of experiments designed to assess force reduction and leash length control, as described in Section III-C.

A. Experimental results

Fig. 20 shows the results of Experiment #1. When a spring is fixed at the end of the rope (blue curve), the peak force

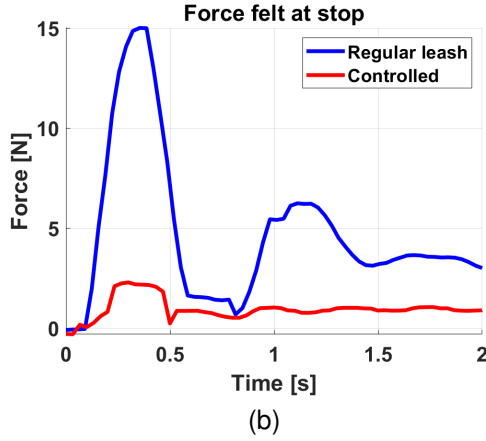
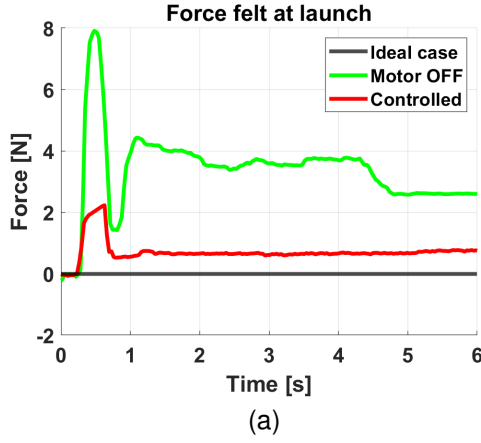


Fig. 19. Results of Experiment #2: measured force during a dog launch (a) and a dog stop (b) with a maximum acceleration of 3.5 m/s^2 , using a spring attached to the end of the leash. The controlled device reduces the sudden impulse during launch or stop.

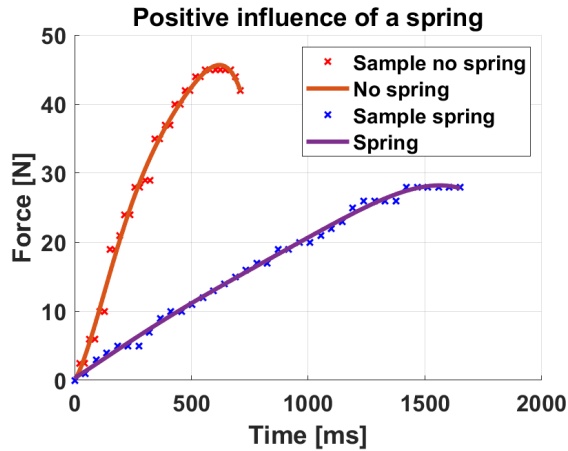


Fig. 20. Results of Experiment #1: measured force after free fall without a spring (red) and with a spring attached at the end of the rope (blue). The use of a spring reduces the peak force and increases the time required to reach the maximum value.

decreases from 46 N to 28 N, corresponding to a reduction of approximately 40%. In addition, the maximum force is reached after 1500 ms instead of 600 ms, indicating that the spring introduces a damping effect. Consequently, the time required to reach the peak force is increased by a factor of 2.5.

Fig. 19 shows the results of Experiment #2. During a dog launch (a), the force experienced by the user reaches 7.9 N when the device is off, compared to 2.2 N when it is active. The maximum force is thus reduced by approximately 75% through the use of the winding motor inertia controller, within its operational limits. With most measured values close to 0.6 N, the prototype performs well at giving a free sensation, even if it not completely free (ideal case at 0N).

In the case of a sudden stop of the dog (b), i.e., when the dog abruptly stops while the user continues moving forward, forces were compared between a regular leash (rope only) and the prototype. With a regular leash, the user experiences a maximum force of 15 N, whereas the prototype limits this force to 2.3 N. This corresponds to a reduction of approximately 83% in maximum force during a full stop.

Fig. 21 shows the results of Experiment #3. This experiment shows the evolution of the leash length (or dog position) over time. Experiment starts with the dog aside of the owner (below the leash length maximum target) and with no speed difference. A rolling mass along a tilted plane (20° slope for an acceleration of $3[m/s^2]$) simulate the dog random behavior. The dog goes through the different state machine of ALIVE, free, braking, forced rewind and braking again. The forced rewind reduce the leash length slightly under the target with a small buffer to avoid oscillation between free and braking. In Experiment #3, the rolling mass doesn't go through the free rewind state since the gravity pull the mass along the slope and maintain a tension in the leash.

Figure 22 shows that adding the spring had a negative impact on the position estimation based on motor feedback. Without the spring, the maximum position error was approximately 15 cm and was solely due to rope overlapping when wound around the spool. However, when the spring was added, this error increased to about 25 cm, mainly because of the spring's extension and contraction, which are unknown to the system.

B. Results improvements

To improve the experimental results, several aspects of the setup and data processing should be refined. First, the plank should be rigidly fixed to the table to prevent unwanted motion. The dynamometer should be more accurate and rigidly mounted on the plank; only an analog dynamometer was available for the experiments, as the digital version exhibited excessive measurement noise. In addition, both cameras should be mounted on a stiff structure to avoid disturbances caused by vibrations induced during the experiments. Second, video-based post-processing was performed using the open-source software *Tracker*, which is commonly used for object position tracking. Two main limitations were identified: insufficient contrast between the tracked marker and the background, and limited tracking accuracy. Developing a dedicated tracking algorithm would allow more accurate position estimation and improved measurement reliability.

VIII. DISCUSSION: PERFORMANCE EVALUATION

As shown in Tab. V, the proposed prototype partially meets the target specifications. Overall, the main limitations arise

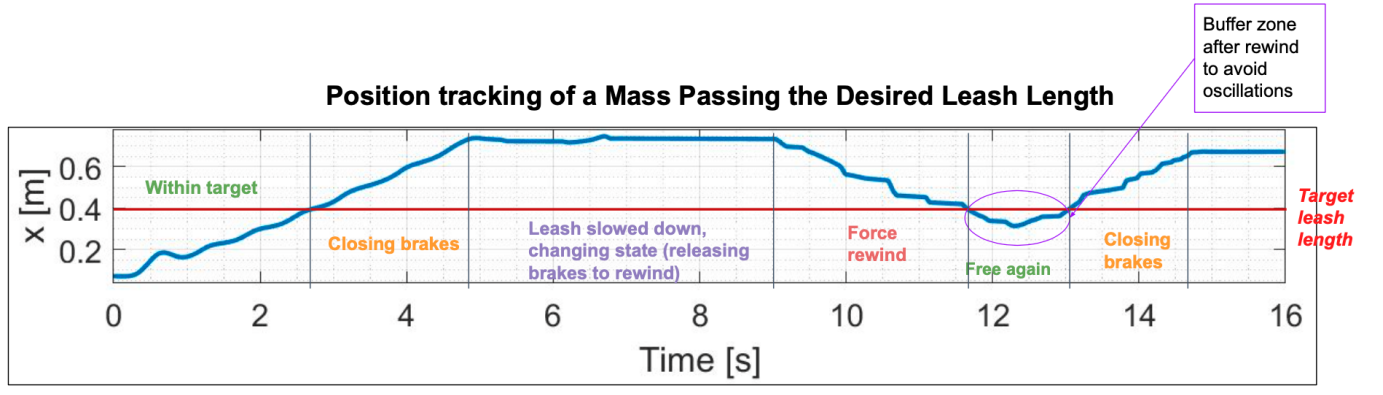


Fig. 21. Results of Experiment #3: Leash length over time, simulated using a rolling mass along a plane with a slope of 20° (acceleration of $3[m/s^2]$). The rolling mass goes through different states (explained on the figure).

TABLE V
SPECIFICATIONS OF TARGET VS PROTOTYPE

Specification	Target	Prototype
Damping range	50%	$\sim 80\%$
Dog specifications	All types of dogs	Small / Slow dogs ($< 10\text{ kg}$, relative speed $< 0.25\text{ m/s}$)
Winding force	150 N	112 N
Braking force	400 N	2000 N
Relative speed	5 m/s	0.25 m/s (below 3 m/s^2 acceleration), fine if dog moves at user speed
Distance of reference	1–2 m	Min 0.5 m / Max 2.5 m

IX. CONCLUSION

This report presented ALIVE, an active leash prototype based on a dual-motor architecture capable of controlling both leash length and pull force. The experimental results show that ALIVE meets its main objective and performs well within the practical constraints of the current prototype (motor speed/acceleration limits). Within this working range, the system reduces peak leash forces and improves comfort compared to a regular leash. The proposed solution combines a winding spool and a brake disc mounted on the same shaft, actuated by a secondary motor through a pinch-based mechanism. While this design validates the feasibility of active stiffness reduction, the current prototype remains limited in terms of size, weight, and mechanical efficiency. Nevertheless, ALIVE serves as a functional proof of concept, providing a solid foundation for future improvements and broader applications of stiffness reduction technologies.

A. Possibility of Improvements

Future iterations of the proposed prototype could focus on downsizing to enhance wearability and ergonomics for active users. A more compact casing would reduce perceived bulk and improve user comfort. As the current prototype weighs 1.688 kg, further weight reduction is also required. This could be achieved through miniaturization of the electronic circuits, for instance by integrating dedicated microcontrollers, which were excluded in the present prototype due to budget constraints.

Regarding the winding mechanism, replacing the current servomotor (initially selected for cost considerations) with

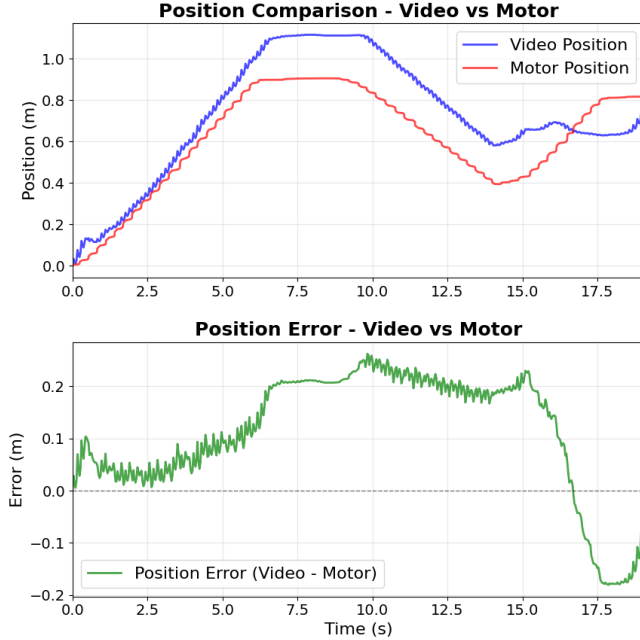


Fig. 22. Results of Experiment #4: Comparison of the ground truth leash length and estimated length (with motor feedback) over time. Simulated using a rolling mass along a plane with a slope of 20° (acceleration of $3[m/s^2]$). The rolling mass goes through different states (free, braking and rewind).

from the limited speed of the winding motor and the inability to handle sudden pulls at high accelerations. These aspects are discussed as key points for improvement in the conclusion.

a backdrivable clutch-based system or a brushless DC motor could significantly improve performance. Brushless DC motors offer reduced friction, increased lifespan, and higher rotational speed, with a trade-off in available torque. Finally, dog motion control could be enhanced by incorporating a guiding or railing mechanism to redirect leash tension when the dog changes position relative to the user.

B. Applications of the proposed technology and future impact

The stiffness reduction mechanism demonstrated in the proposed prototype could be applied to canine leashes in its current form. However, for use by everyday users, further improvements would be required. Other potential application areas include wearable safety devices for humans. For example, climbing equipment could integrate such mechanisms to limit shock loads in the event of a fall while giving room to move around, thereby reducing the risk of injury. Similarly, safety harnesses used by workers could benefit from an adaptation of this technology. These examples illustrate the broad potential impact of the proposed technology across multiple fields.

X. APPENDIX

A. Cost Analysis

Description	Cost Budget		
	Cost [CHF]	Cost contingency	Estimated Cost [CHF]
Orders	147,30	0%	147,30
3D printing parts	29,60	0%	29,60
Laser Cutting parts	0,00	0%	0,00
Metallic parts	9,00	0%	9,00
Nuts, screws and inserts	8,50	0%	8,50
Cables and electronics	5,00	0%	5,00
Total	Cost [CHF] :		199,40
Prototyping	15,00	0%	15,00
Total with prototyping	Cost [CHF] :		214,40

Fig. 23. Cost budget for the design, showing the final device cost (199.40 CHF) and the total project cost (214.40 CHF).

Fig. 23 shows the cost analysis of the proposed prototype, which includes both the final device and the overall development process. The final device costs 199.40 CHF and corresponds to the components integrated into the prototype directly, such as both motors, mechanical parts, electronic components, and the casing. The total project cost reaches 214.40 CHF, accounting for additional expenses related to prototyping, including spare parts, testing materials, and iterative development. This cost breakdown highlights that the majority of the budget is allocated to functional components, while the development overhead remains limited.

B. Gantt chart

Fig. 24 provides an overview of the prototyping timeline. It summarizes all major tasks from project initiation to report submission.

C. Division of the work

The work was distributed according to each member's expertise to optimize the development of the final prototype. Two members focused on software-related tasks, including control design and implementation, while the remaining four members were responsible for hardware aspects such as calculations, prototyping, and assembly. The main tasks were summarized in Tab. VI

D. CAD design

See below

ACKNOWLEDGMENTS

The authors would like to thank the teaching assistants of the Reconfigurable Robotics Lab for their valuable guidance and constructive feedback throughout the semester. They also gratefully acknowledge SPOT for providing access to prototyping tools and practical advice during the development of the prototype.

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XI. NOTICE ON AI TOOLS

All concepts, analyses, designs, and results presented in this report are the result of the authors' work conducted throughout the semester. AI tools were used solely to assist with English language correction and stylistic improvements, as the authors are non-native English speakers. The use of these tools did not contribute to the generation of experimental results, which remain entirely the authors' original work.

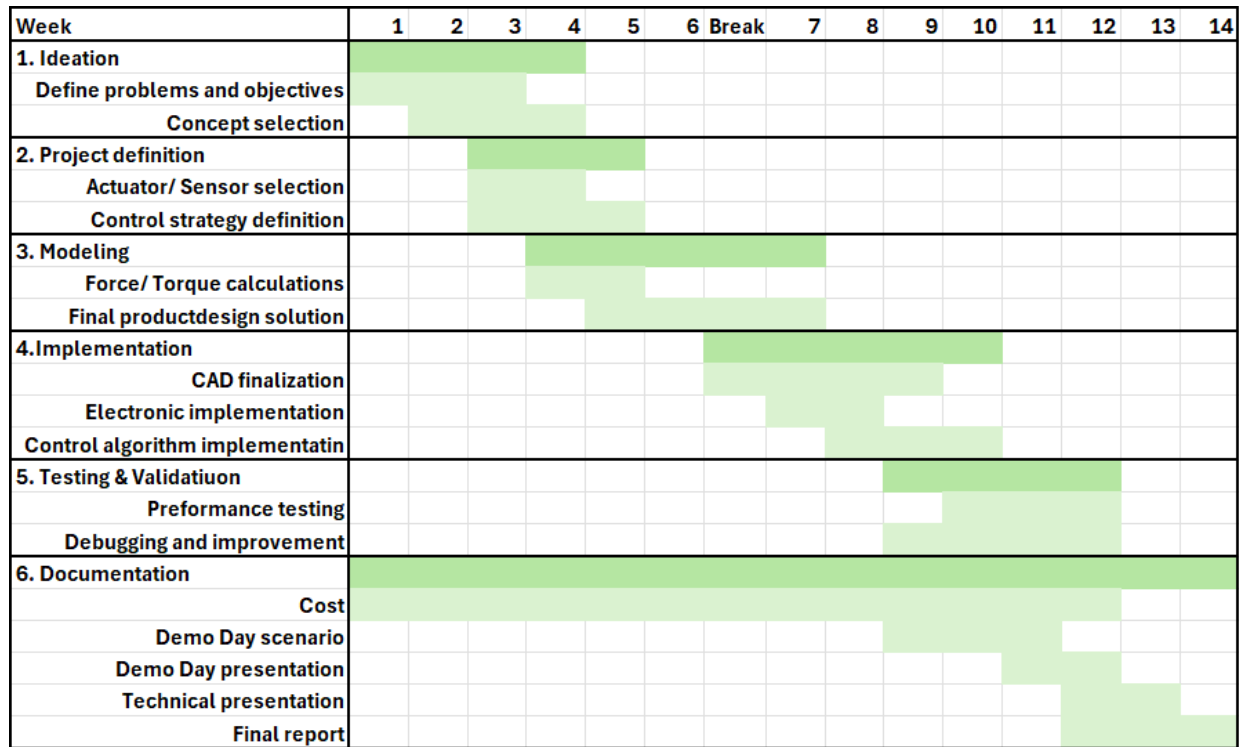


Fig. 24. Gantt Chart for the A.L.I.V.E. project. Main sections were divided into subsections for a more accurate reading.

TABLE VI
DIVISION OF WORK AMONG TEAM MEMBERS

Members	Division	Main Tasks
Andrea	Software	Control design (brakes PID and calibration of brakes), coding, system integration, calculation, debugging, testing, hardware
Aude-Line	Software	Control design (data acquisition, FSM, free state controller), coding, electrical design, system integration, hardware fixing, testing
Bastien	Hardware	Mechanical calculations, prototyping, testing, housing and belt fixation, data recovering
Zoé	Hardware	Mechanical calculations, prototyping (passive spring system - discarded), testing, data recovering
Alfonso	Hardware	Brake mechanism design, manufacturing
Vincente	Hardware	Winding mechanism design, housing structure, manufacturing, mechanical calculation, electrical design, data monitoring code

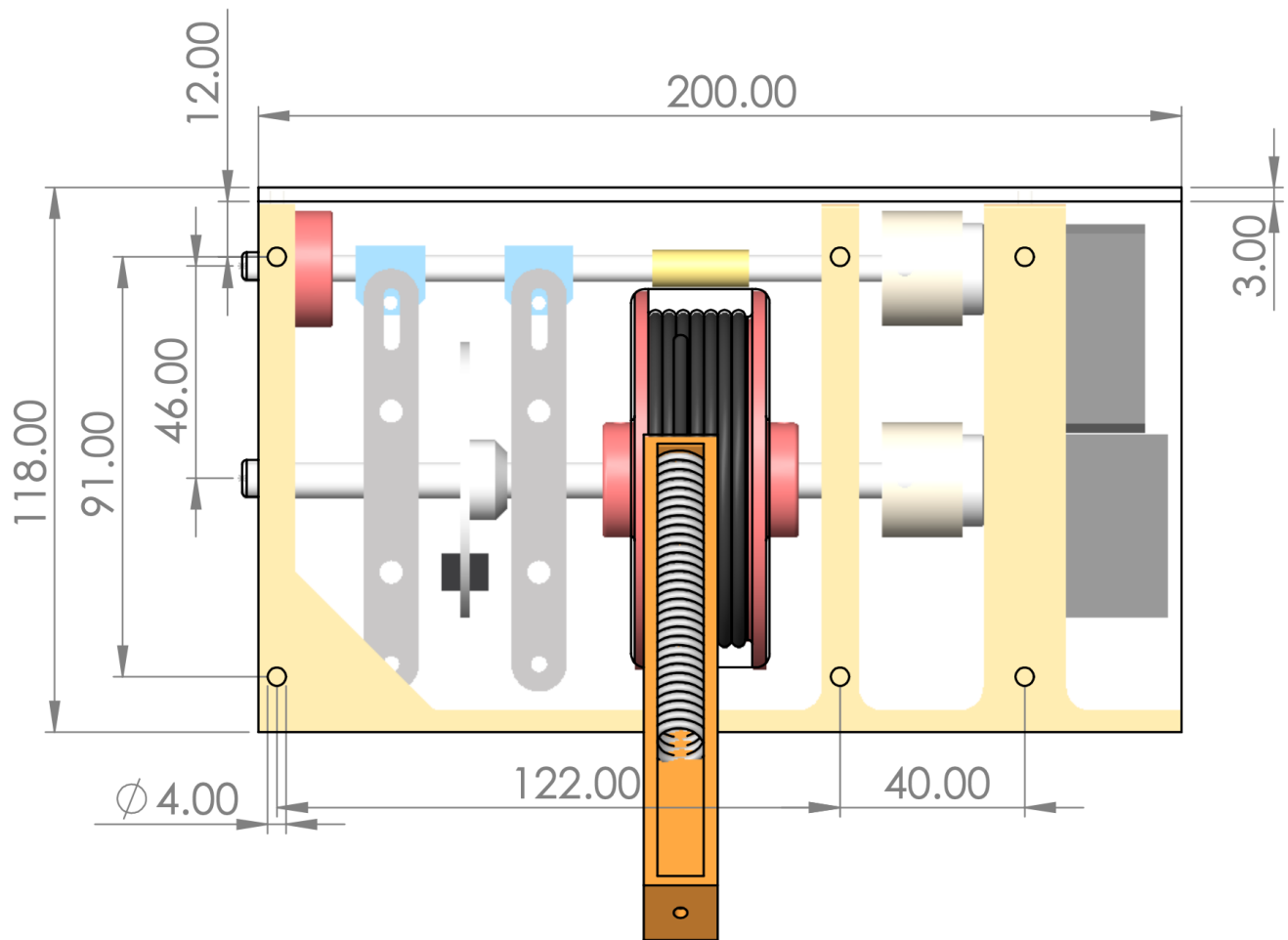


Fig. 25. Face view of the overall design and dimensions

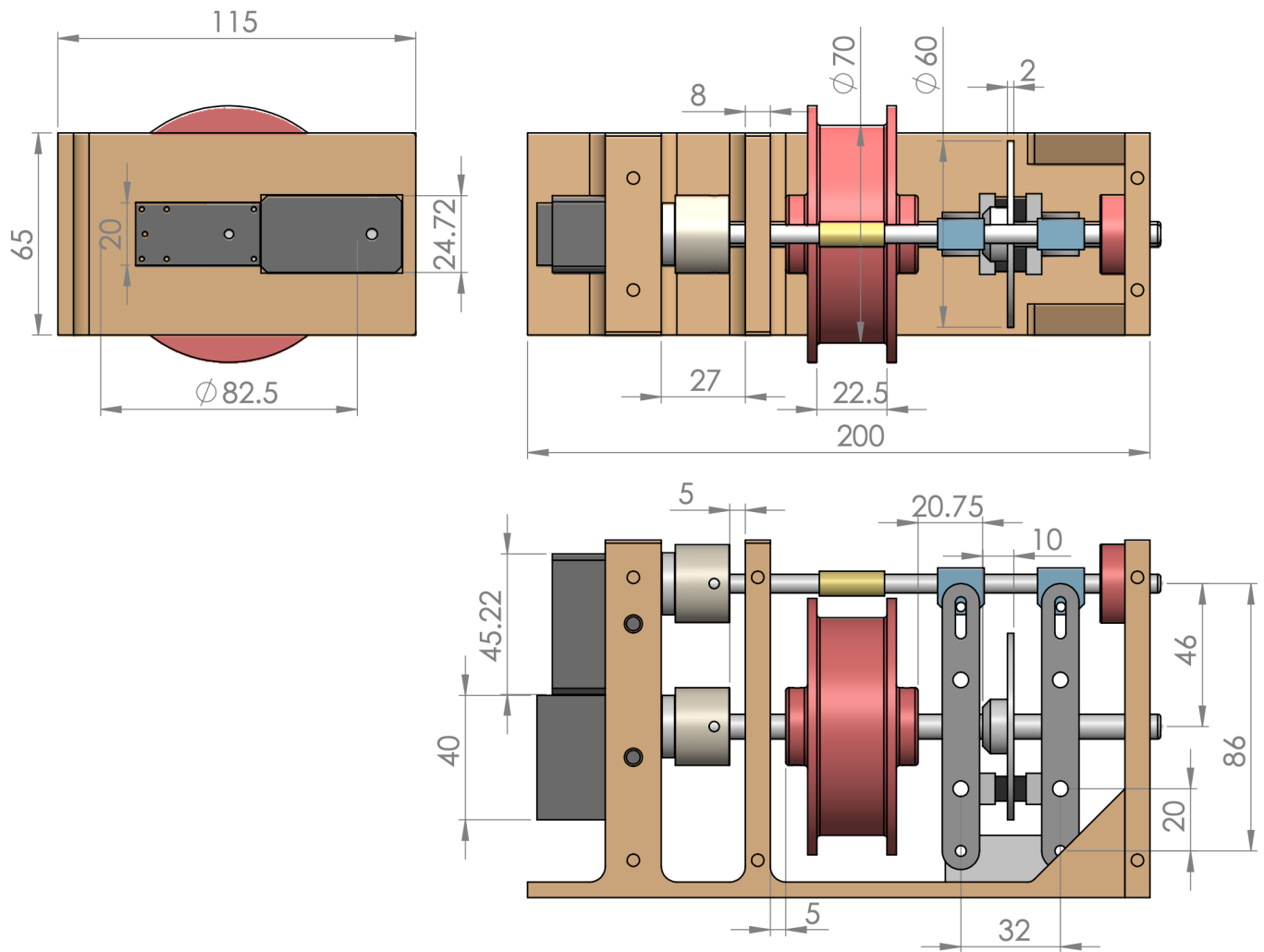


Fig. 26. Main dimensions of ALIVE

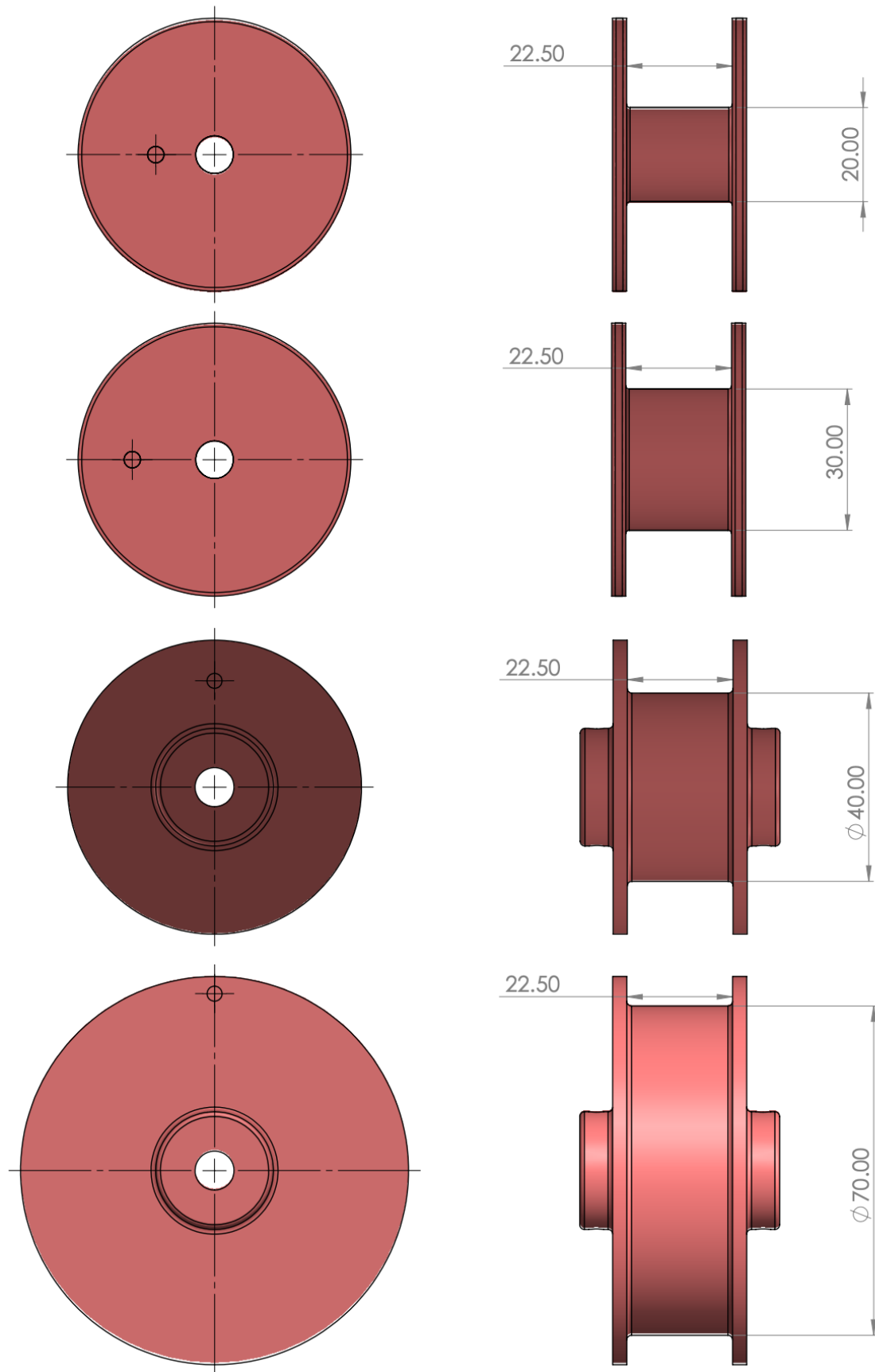


Fig. 27. Spool diameter evolution along the project

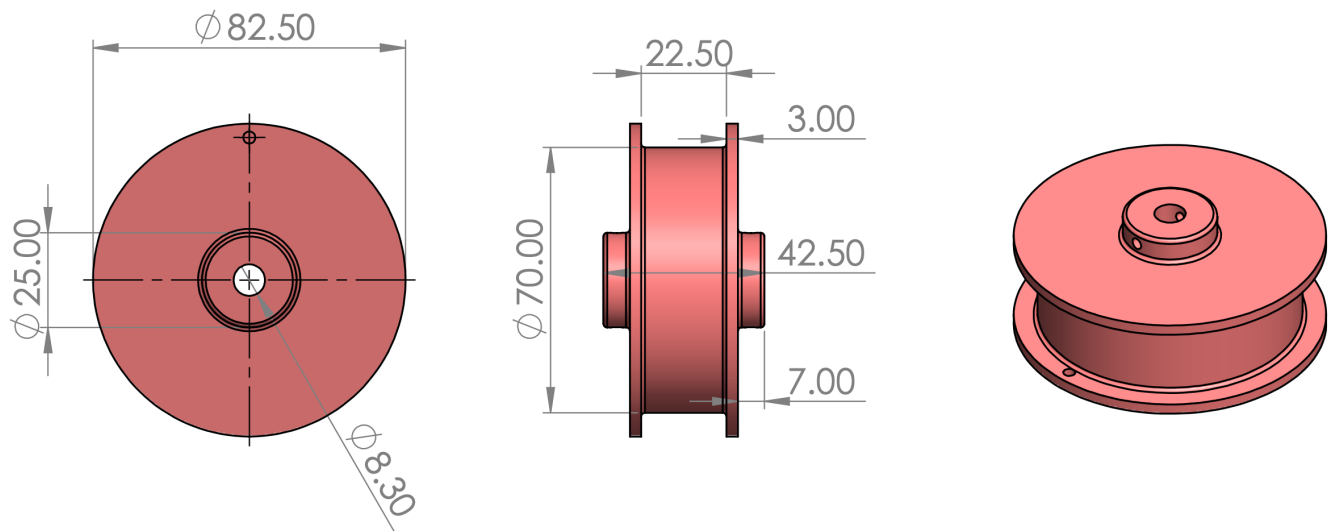


Fig. 28. Spool dimensions

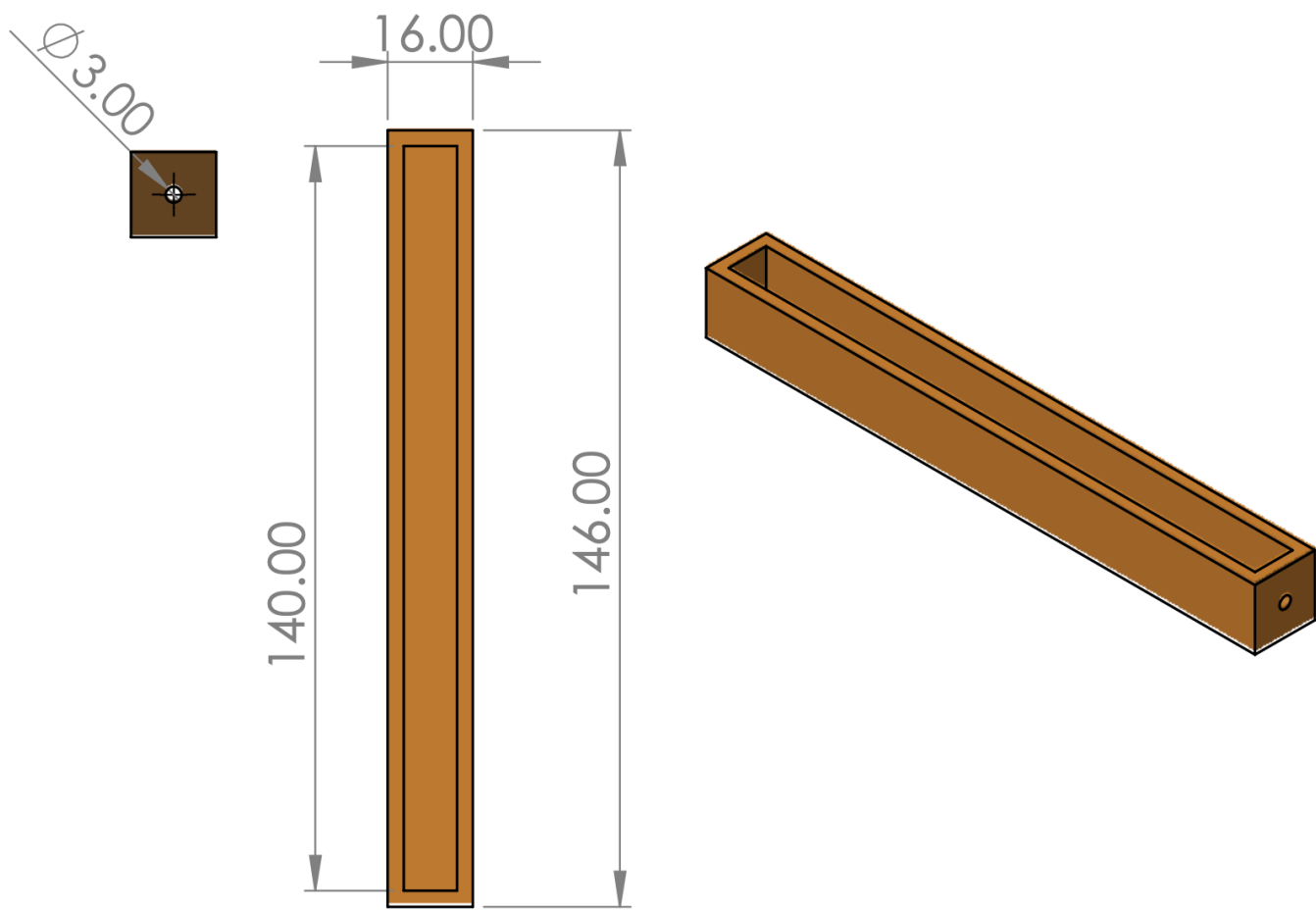


Fig. 29. Spring housing that limit the elongation of the spring to avoid plastic deformation or rupture.